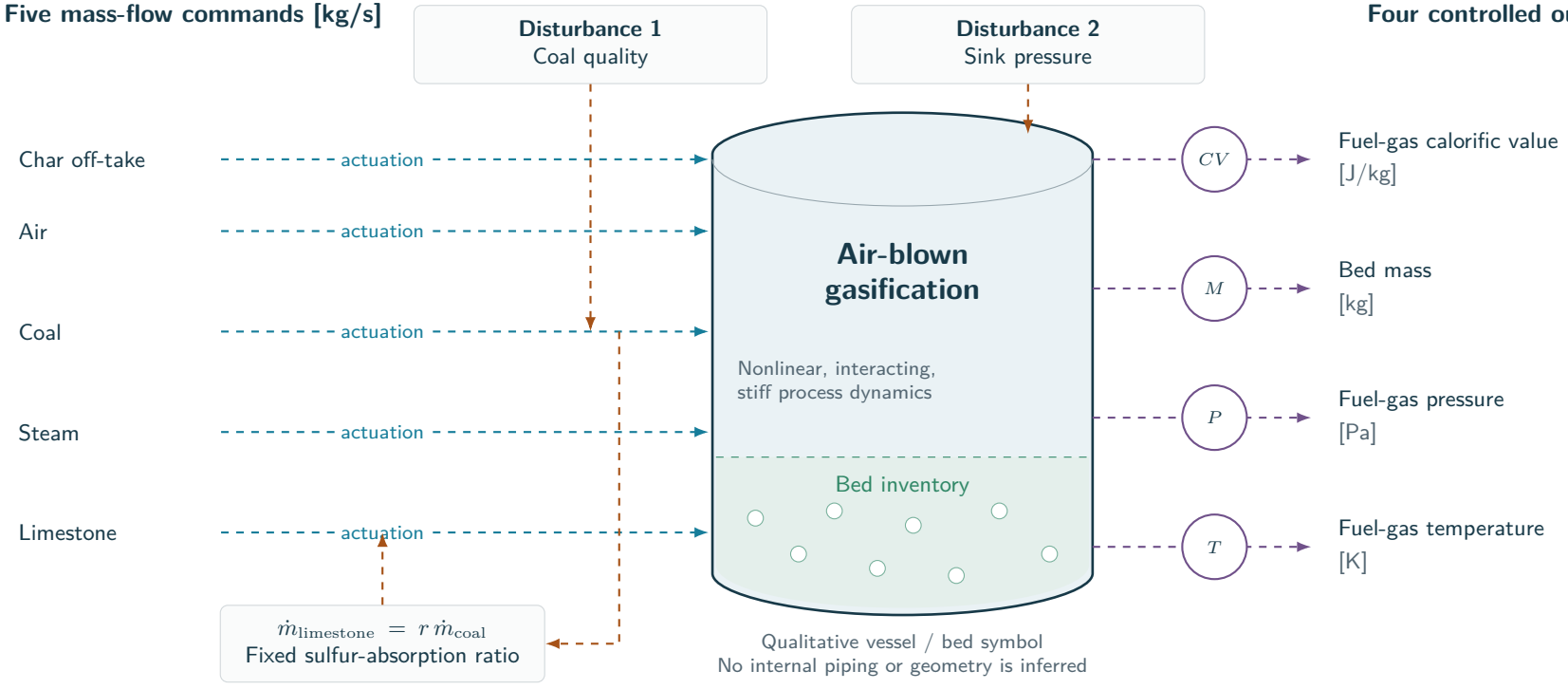
ALSTOM gasifier: nonlinear plant and control interface

Paper section 3.11: *ALSTOM Gasifier Control Benchmark II*



Four independent manipulated flows; the fifth follows coal. Magnitude and slew limits apply to the actuators.

Official model family

Nonlinear open-loop plant and multivariable PI baseline. The native package contains the dynamics; this drawing specifies the external signal interface.

Challenge experiments

Sink-pressure steps and harmonics; load and coal-quality changes. Compare maximum and integral absolute output errors with the declared bounds.

Interpretation of results

A completed simulation is not a passed control specification. The retained native PI results exceed the temperature band in all six pressure cases.

Dashed blue arrows: mass-flow control signals, not physical pipe directions; char is withdrawn from the process. Orange: disturbances and fixed-ratio linkage. Purple: output measurements, not material outlets. Vessel and bed are conceptual; this is not a construction P&ID.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.